

Energy-Aware Arm and Base Selection for a Ceiling-Rail Dual-Arm Robot

Using a Base-Aware GNG Capability Map

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Motivation: service robots in the home

- Service robots increasingly fetch and tidy in cluttered homes shared with people [Lisondra2025, IFR2025].
- Floor-standing manipulators occupy walkable space and raise ground-level safety concerns.
- Lift the robot off the floor: arms hang from an overhead rail and approach objects *from above* [Jayathilaka2025].
- Our platform: **two arms on one sliding & rotating ceiling rail.**

*Fig. 1 (left): ceiling-rail
dual-arm platform
TODO: add asset*

Problem: which arm, and where to place the base?

- The rail moves both arms, so each arm + rail is an **8-DOF redundant chain**:

$$q = [d, \theta, q_1, \dots, q_6]$$

with d the rail translation and θ the rail rotation.

- An object is usually reachable by *either* arm, from *many* rail positions.
- Two coupled decisions at once: **which arm** and **where the base goes** before reaching.
- Repositioning the heavy base dominates time and energy cost [MoMaPos2024, BaSeNet2024, EnergySLR2024].

Gap: capability maps are arm-only and binary

- Classic reachability / capability maps assume a **fixed base** [Zacharias2007, Makhal2018].
- They record only *whether* a cell is reachable — no reaching cost, no IK seed.
- Base-placement methods pick *where one arm stands*, not *which of several arms* under an energy budget [MoMaPos2024, InvReach2022].

Missing

A representation that answers, directly, **which arm and which base placement cost the least energy** on a redundant ceiling-rail robot.

Related work: what is different here

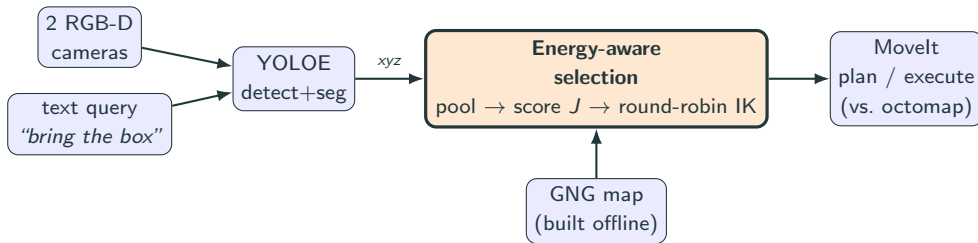
Line of work	Representative	How ours differs
Reachability / capability maps	Zacharias'07, Reuleaux [Makhal2018], RM4D2024	node carries $q + \text{manip} + \text{hold}$ (not a binary voxel); the rail is part of the configuration
Base placement (mobile manip.)	MoMa-Pos2024, BaSeNet2024, Inv-Reach2022	choose <i>which arm</i> + base by energy , not where one arm stands
Redundancy / inverse kinematics	Yoshikawa'85, IKDif-fuser2025	map supplies the IK seed; arm + base settled by the cost, IK stays lightweight
Dual-arm allocation	DAG-Plan2024, RoboPARA2025	allocate by <i>energy</i> for one reach, not a task / temporal graph

System: ceiling-rail dual-arm + two RGB-D cameras

- 2× Kinova Gen3 Lite (6-DOF, ~ 0.76 m reach) on a shared rail [Kinova].
- **Sim-only** in NVIDIA Isaac Sim; same URDF as the hardware [IsaacSim2026].
- Two overhead RGB-D cameras: color $\rightarrow$ YOLOE detect, depth $\rightarrow$ collision octomap [Wang2025].
- Pipeline: perception $\rightarrow$ energy selection $\rightarrow$ MoveIt plan/execute [Coleman2014].

*Fig. 1: system + frames —
end-to-end pipeline
TODO: add asset*

End-to-end pipeline



Perception is an *enabler*; the **energy-aware selection** (orange) is the contribution. Depth from the *same* two cameras builds the collision octomap that Movelt plans against. (Fig. 1, right panel.)

Perception input: an open-vocabulary text query

The operator names the object in free-form text; command / filler words are stripped and the remaining noun becomes the YOLOE prompt.

Example queries

- “bring the banana”
- “pick up the teddy bear”
- “get me the yellow box”
- “hand me the can”
- “can you fetch the bottle”

Understood as

- verbs *bring / get / grab / pick / hand / fetch / take* are dropped $\Rightarrow$ object noun remains.
- **colour-aware:** “yellow box” $\rightarrow$ class box, colour yellow.
- **aliases:** “doll” = “teddy bear”; “tin can” = “can”.

Enabler only: the query yields the target position that the capability map and cost J then act on [Wang2025].

Base-aware GNG capability map

One map per arm; nodes tile the reachable workspace [Fritzke1995]. **Each node stores:**

Sym.	Contents	Dim.
x	end-effector position (task part)	3
q	full 8-DOF config $[d, \theta, q_1 \dots q_6]$ (IK seed)	8
w	manipulability $\sqrt{\det(JJ^T)}$	1
h	holding cost $\ gravity\ torque\ $	1

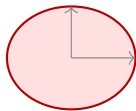
- BMU search uses x (tile workspace); q is a ready IK seed — config *and* rail, not only quality values.
- Rail DOF (d, θ) sampled $\Rightarrow$ map captures reach that **base motion** adds.
- Boundary shell *pinned* to the true reach surface.

Fig. 4: GNG map coloured by manipulability, pinned boundary shell
TODO: add asset

Manipulability: how dexterous is a posture?

$$w = \sqrt{\det(JJ^T)} \quad [\text{Yoshikawa1985}]$$

- J — the arm's **Jacobian** at q : $\dot{x} = J\dot{q}$ (here 6×8 : 6-DOF pose, 8-DOF chain).
 - JJ^T collapses the rectangular J into a 6×6 matrix we can take a determinant of.
 - w is the **volume of the manipulability ellipsoid** ($w = \sigma_1 \cdots \sigma_6$, the singular values of J).
- **large** w : round ellipsoid — moves easily in any direction.
- **small** $w \rightarrow 0$: collapsed — near a **singularity**, IK ill-conditioned.



high w (dexterous)



low $w \rightarrow 0$ (singular)

In this work: w is a per-node map layer (Fig. 4) and enters cost J with a *minus* sign, so a more

Energy cost J

$$J = w_{gl} \frac{\Delta_{lin}}{\rho_{gl}} + w_{gr} \frac{\Delta_{rot}}{\rho_{gr}} + w_{arm} \frac{\Delta_{arm}}{\rho_{arm}} + w_{dist} \frac{e}{\rho_{dist}} + w_{hold} \frac{h}{\rho_{hold}} - w_{manip} \frac{m}{\rho_{manip}}$$

- Rail-linear, rail-rotation, arm travel; tool-to-object distance e ; holding torque h ; $-$ manipulability m .
- Rail linear & rotation carry *separate* weights (metres vs radians of a heavy carriage).
- Each term $\div \rho$ (median over calibration picks) $\Rightarrow$ dimensionless; a weight reads as a priority.
- Proxy for pick energy — not exact energy; validated against measured trajectory energy [EnergySLR2024].

Term	Symbol	Weight	Ref. ρ
Rail linear travel	Δ_{lin}	2	0.95
Rail rotation travel	Δ_{rot}	12	0.70
Arm joint travel	Δ_{arm}	20	6.00
Tool-to-object distance	e	3	1.36
Gravity holding torque	h	1	2.90
Manipulability (subtr.)	m	1	0.145

Table 1. Calibrated weights and references for J .

Selection: pooling and round-robin search

1. **Pool** — gather map nodes within a radius of the object, from *both* arms' maps (the reachability filter).
2. **Score** every candidate by J ; both arms compared on one scale.
3. **Round-robin** — interleave the two arms so one cannot consume every attempt; lowest- J arm gets a small head start.
4. **Plan-only** — seed IK from the node, ask MoveIt for a collision-free plan; the arm never moves during the search.
5. **First valid plan wins** — only that candidate is executed; on abort, fall through to the next.

*Fig. 5: J picks the cheaper arm/base (may not be the nearest)
TODO: add asset*

Results (NVIDIA Isaac Sim)

- (a) Treating the rail as a DOF enlarges the reachable workspace $\sim 5.1\times$ vs. an arm-only baseline (E1).
- (b) GNG-seeded IK: success $\uparrow$, solve-time $\downarrow$ — **[results pending]**.
- (c) Energy-aware vs. nearest / fixed / random: base travel and mechanical energy — **[results pending]**.

All numbers come from Isaac Sim; physical validation is future work.

*Fig. 6: (a) workspace gain $\sim 5.1\times$
— (b) energy vs. baselines
[pending]
TODO: add asset*

The map + energy cost J extend naturally along several directions:

- **Dual-arm collaboration.** Use *both* arms together — for heavy objects or carrying more than one item at once — scored on the same shared-base energy cost [DAGPlan2024, RoboPARA2025].
- **Object hand-over.** Deliver to a person (the “hand things to the people” goal of the intro), and *arm-to-arm* hand-over whose transfer point is chosen by the same cost J .
- **Human–robot interaction.** Read human pose / pointing gestures to disambiguate *which* object is wanted, and to reach around people safely [Lisondra2025].
- **Contact-based grasping.** Move from the current approach-only, plan-only search to real grasp execution (contact, close, lift) with regrasp on failure.
- **Energy-aware multi-pick sequencing.** Extend J from one action to a *sequence* of picks, planning an energy-optimal base-pose order [BaSeNet2024].

Toward real-world deployment

- **The platform already exists.** Four Kinova Gen3 Lite arms, a Modbus-driven ceiling gantry, and a Livox Mid360 LIDAR — driven from the *same* URDF used in simulation.
- **Direct energy validation.** Measure real rail + arm motor current / power to confirm that minimizing J lowers *measured* energy — backing the “Energy-Aware” claim beyond the simulated proxy.
- **Perception likely improves.** YOLOE is trained on natural images, so real RGB-D should be *more* reliable than synthetic Isaac renders.
- **Open gaps.** Heavy-rail dynamics and friction, kinematic calibration, and safety while sharing space with people.

- **Contributions:**

- Energy cost J that selects the arm *and* the full 8-DOF configuration together (rail travel included).
- A base-aware GNG capability map: per-node IK seed, manipulability, and holding cost, with a pinned boundary shell.
- Evaluation in Isaac Sim vs. nearest / fixed / random baselines.
- Treating the ceiling rail as a real DOF pays off: $\sim 5.1\times$ larger reachable workspace.
- **Sim-only** here $\rightarrow$ hardware validation on the physical platform is left for future work.

Thank you — questions?

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